

AI-POWERED CAD ANALYSIS REPORT

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Geometry to Process to Cost · Fully AI-reasoned

60

Automotive HDPE Fuel Tank (EBM, multilayer barrier)

1528 × 658 × 594 mm · 10595.9 cm³ · AI 28.609 kg / Steel 83.178 kg

Manufacturability: 60/100 · Confidence: Low

§1 — Geometry & Part Summary

Bounding Box	1528.3 × 657.6 × 593.7 mm	Surface Area	47571.3 cm ²
Volume	10595.90 cm ³	Weight (AI)	28.609 kg
Weight (Steel)	83.178 kg	Weight (Plastic)	11.126 kg

§2 — Detected Features

Feature Type	Count	Significance	Description
Cylindrical faces (pipe stubs/vent tubes/bosses)	1375	High	1375 cylindrical faces, radii 1-42mm, consistent with moulded-in fill/vent/mounting stubs on a fuel tank

§3 — Material Analysis

AI-suggested

Primary Material: Unspecified (50% confidence)
undefined

§4 — Process Recommendations

Process	Commodity	Confidence	Cycle Time (hr)	Reasoning
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§7 — AI Indicative Cost Range (model opinion — not engine-calculated)

Net Weight	—	Material	—
Recommended Process	machining	Cycle Time	—

Setup Time	—	Operations	0 ops
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Operations Detail

§8 — AI Analysis Explanation